American Computer Science League

2024-2025 ● Contest 1: Short Problems ● Intermediate Division

1. Computer Number Systems

Convert 2024_{16} to base 2.

A. 10000001000100
B. 10000000100100
C. 10000000010100
D. 10010010000000
E. 10010000000100

2. Computer Number Systems

Evaluate and express the answer in octal.

$$4265_8 + 6524_8$$

A. 10701
B. 10789
C. 11001
D. 13011
E. 13110

3. Recursive Functions

Find $f(5)$ given:

$$f(x) = \begin{cases} 256 & \text{if } x \leq 1 \\ \dfrac{f(x-1)}{2} & \text{if } x > 1 \end{cases}$$

A. 128
B. 48
C. 8
D. 100
E. 16

4. Recursive Functions

Find $f(12, 10)$ given:

$$f(x,y) = \begin{cases} f(x-3, y+1) - 2 & \text{if } x > y \\ f(x+1, y-4) + 1 & \text{if } x < y \\ 2x & \text{if } x = y \end{cases}$$

A. 6
B. 7
C. 8
D. 9
E. 10

5. What Does This Program Do?

What is the output of the following program?

```
h = 200
y = int(h / 36)
h = h % 36
f = int(h / 12)
i = h % 12
output y, f, i
```

A. 5, 200, 8
B. 5, 20, 1
C. 5, 1, 8
D. 5, 1, 20
E. 5, 8, 1

6. What Does This Program Do?

What is output by the following program?

```
s = 0
for x = 1 to 5
     s = s + x ^ 2
next x
for y = 2 to 10 step 2
     s = s + y / 2
next y
for z = 5 to 1 step -1
     s = s - z * 2
next z
output s
```

A. 40
B. 55
C. 70
D. 30
E. 180